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(54) Title of the invention : TRANSFORMER-BASED MODEL LEVERAGING CUSTOMIZED AUTOREGRESSION ON SEMANTIC WORD INSIGHT

(51) International classification :G06F0040300000, G06N0020000000, G06F0016350000, G06N0003045000, G06F0040284000 (86) International Application No :NA Filing Date :NA (87) International Publication No : NA (61) Patent of Addition to Application Number :NA Filing Date :NA (62) Divisional to Application Number :NA Filing Date :NA		(71)Name of Applicant : 1)Manipal University Jaipur Address of Applicant :Manipal University Jaipur, Off Jaipur-Ajmer Expressway, Post: Dehmi Kalan, Jaipur-303007, Rajasthan, India Jaipur ----- Name of Applicant : NA Address of Applicant : NA (72)Name of Inventor : 1)Dr. Priya Goyal Address of Applicant :Department of Artificial Intelligence & Machine Learning, Manipal University Jaipur, Jaipur-Ajmer Express Highway, Dehmi Kalan, Near GVK Toll Plaza, Jaipur, Rajasthan, India 303007. Jaipur ----- 2)Siddhanth Bhat Address of Applicant :Department of Artificial Intelligence & Machine Learning, Manipal University Jaipur, Jaipur-Ajmer Express Highway, Dehmi Kalan, Near GVK Toll Plaza, Jaipur, Rajasthan, India 303007. Jaipur ----- 3)Sanya Dureja Address of Applicant :Department of Artificial Intelligence & Machine Learning, Manipal University Jaipur, Jaipur-Ajmer Express Highway, Dehmi Kalan, Near GVK Toll Plaza, Jaipur, Rajasthan, India 303007. Jaipur ----- 4)Prisha Goyal Address of Applicant :Department of Artificial Intelligence & Machine Learning, Manipal University Jaipur, Jaipur-Ajmer Express Highway, Dehmi Kalan, Near GVK Toll Plaza, Jaipur, Rajasthan, India 303007. Jaipur -----
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(57) Abstract :

The present invention relates to a system and method for the semantic categorization of online news articles. The system comprises a dataset loading module that contains the last 10 year BBC All Time and Reuters News; data pre-processing module to enhance data quality through technique such as lemmatization, stemming, tokenization, lowercasing, and stopword removal; Dataset Transformation module to eliminate superfluous columns from the combined BBC All Time and Reuters news databases; and a module to train the data set to classify the news. The present invention's Transformer-Based Model Utilizing Tailored Autoregression on Semantic Word On the English News Dataset, Insight outperforms these general models, achieving better classification results, especially when it comes to news title classification. This transformer-based model can serve as the foundation for next forecasts and investigations into the processing of news datasets, according to a comparison with cutting-edge research. This transformer-based model achieves a state-of-the-art 96.19% and 82.7% training and validation accuracy respectively.

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